Research Question Sharpener

Project: Phylogenetic Tree Recovery from Representation Geometry in Multilingual Language Models

SECTION 1: THE WHAT

What are you claiming or showing?

We measure whether the residual-stream geometry of multilingual language models contains phylogenetic signal sufficient to recover the genealogical tree topology of related language families. For each pair of languages in a family, we fit a linear transformation between their contextual representations and define *composition error* as the discrepancy between a composed indirect transformation (A→B→C) and the direct transformation (A→C). We then ask: which tree topology over the family minimizes total composition error across all triples? We compare the recovered topology to the established phylogeny.

This is a measurement, not an existence claim. The correct Romance tree is not controversial. The chance baseline is calculable (1 in 15 for five taxa). Either the geometry recovers the tree or it does not. We report results across three language families (Romance, Germanic, Dravidian) and multiple model architectures. We further characterize the signal by varying the degree of language-identification projection and by testing whether areally proximate but genealogically distant languages are correctly placed as outgroups.

What the project does *not* claim: we do not claim the model "knows" historical linguistics, that it has recovered a proto-language, or that a specific theoretical framework explains the result. We report a geometric measurement and characterize what it contains.

The Surprise

The standard account of cross-lingual geometric structure in multilingual models posits two components: a coarse language-identification signal and surface-similarity-driven typological clustering. Under this account, Tamil and Kannada are close in representation space because they share script, lexicon, and typological features. The surprise is that a third component, carrying phylogenetic branching-order information, is present and separable. Knowing that Spanish and Portuguese are similar does not tell you whether Romanian branches off before or after the Italo-Western split. Getting the internal branching right requires structural signal that goes beyond pairwise similarity.

Who updates: multilingual NLP researchers whose working model of cross-lingual representation does not include a separable phylogenetic layer. Computational historical linguists, because a genuinely new data source for phylogenetic inference has not appeared in a long time. Mechanistic interpretability researchers, because the finding is about transformation structure between representations rather than about what individual features encode.

One-Sentence Version

The tree topology that minimizes composition error of pairwise linear transformations between related-language representations in multilingual LLMs recovers established phylogenies for Romance, Germanic, and Dravidian, and this signal is separable from generic language identification.

Alignment with LossFunk

Primarily Direction 2 (Creativity in artificial systems). LossFunk argues that creativity requires structured representations of possibility spaces. If the model's geometry encodes not just which languages exist but the branching structure connecting them, it has built a parameterized space of linguistic possibilities in which points not visited during training (ancestral states, hypothetical intermediates) are geometrically locatable. Whether this space can be traversed generatively is a downstream question; whether it exists is what we test. The project also connects to Direction 3 (Biological foundations of intelligence): languages evolve like species, and phylogenetic inference from learned representations tests whether neural networks discover the transformation operators connecting related instances of a structured domain.

SECTION 2: THE WHY

New questions this result would open

1. Domain generalization. If representation geometry encodes phylogenetic structure for languages, does the same hold for other domains where related instances share generative history? Biological phylogeny from protein-language models, art-historical periodization from vision models, musical genre trees from audio models. The method (composition-error-minimizing tree recovery) is domain-agnostic.

2. Signal decomposition. Given that the phylogenetic signal exists, what carries it? Do the principal components of the phylogenetically informative transformations correspond to morphological paradigm shifts, phonological correspondences, syntactic parameter changes, or something the model has learned that does not map onto any traditional linguistic category? This is a standalone follow-up.

3. Threshold and scaling. How much daughter-language training data, and how many sister languages, are required before phylogenetic signal becomes recoverable? Testable by controlled experiments at varying data scales. Tulu (sparse training data, late-branching South Dravidian) is a natural probe of this gradient within the Dravidian family.

4. Contested phylogenies. For families where the correct tree is debated, the geometric tree becomes a new, independent data source. This is interesting regardless of which side it favors, and it is the strongest possible test against the retrieval confound.

Who would build on this?

Computational phylogeneticists (Bouchard-Côté and collaborators at McGill; Bouckaert, Greenhill, and the Bayesian phylogenetics community) gain a new data source for tree inference. Multilingual NLP researchers (AI4Bharat, Sarvam, the NLLB team) get a diagnostic for what their models encode beyond surface similarity. Mechanistic interpretability groups (Anthropic, EleutherAI) get a concrete instance of relational structure in activation geometry. Cognitive scientists working on structural representation learning (Lake, Tenenbaum) get a test case for relational abstraction in neural networks.

Crossroads or corridor?

Crossroads. The result connects phylogenetics (new data source), multilingual NLP (what models encode), mechanistic interpretability (transformation geometry), and the general question of whether neural networks learn relational structure from distributional data. Each community picks up a different thread.

SECTION 3: THE HOW

Killer Alternative Explanations

Threat 1: Language identification. Multilingual models encode a coarse 'which language is this' signal. A naive cross-lingual transformation will be dominated by this component. Compositionality of generic language-switch operations is trivial and carries no phylogenetic information.

What forecloses this: we do not pick a single language-ID projection. We run tree recovery at every projection dimensionality from 0 (no projection) to k (aggressive projection) and report the full curve. If tree recovery improves as language-ID dimensions are removed, peaks at some dimensionality d*, then degrades, this both demonstrates a separable phylogenetic component and locates its dimensionality. No methodological degrees of freedom remain because no projection choice is made. The curve is the result.

Threat 2: Retrieval of linguistic scholarship. The model has read comparative linguistics literature. Tree recovery could reflect memorized classification rather than structural inference from distributional data.

Three responses, ordered by strength. First, retrieval of family-membership labels produces clustering, not directional transformation structure with correct internal branching order. Knowing 'these are all Romance' does not tell you where Romanian branches off. Second, we replicate on families with sparser digital classification literature (selected Bantu and Austronesian subfamilies) and compare recovery accuracy. Third, we test on contested phylogenies where the model cannot have memorized the right answer because linguists disagree. If the geometric tree matches one side of a genuine debate, retrieval is ruled out for that case.

Threat 3: Typological similarity masquerading as genealogy. Related languages tend to be typologically similar. Composition error might track typological distance, not genealogical structure.

What forecloses this: the areal-outgroup test. We include a contact language that is typologically similar to the family but genealogically distant. Marathi for Dravidian (millennia of contact, shared SOV order, shared postpositions, but Indo-Aryan). Maltese for Romance (heavy Sicilian/Italian contact but Semitic). If tree-fitting correctly places these as outgroups, the signal is genealogical. If it embeds them within the family, the signal is typological. We also test typologically similar but unrelated pairs (e.g., Japanese and Tamil: both agglutinative SOV) to verify that composition error does not produce spurious phylogenetic signal.

Threat 4: Compositionality is trivially true. If all transformations are fit on overlapping data, composition will approximately hold as an artifact of consistent fitting.

What forecloses this: strict data separation. T_AB is fit on cognate set X, T_BC on disjoint set Y, T_AC on disjoint set Z. Composition is tested on a fourth held-out set. Additionally, composition error is evaluated against all possible tree topologies, not just the predicted one. If composition holds trivially, all topologies should perform similarly. The phylogenetic claim requires the correct topology to be significantly better than alternatives.

Experimental Design

Pilot (1-2 weeks). The go/no-go gate. Five Romance languages (Spanish, Italian, French, Portuguese, Romanian). 500 parallel sentences from FLORES-200. Mean residual-stream representations extracted at layers {1/4, 1/2, 3/4, final} of a general multilingual model (Gemma-2 or Llama-3). Procrustes maps fit between all 10 pairs. Composition error computed for all 10 triples. Total composition error computed for all 15 possible unrooted topologies on 5 taxa. Report: does the minimum-error topology match the consensus Romance tree? If not, the project ends here. One week, no specialized infrastructure.

Phase 1: Language-ID characterization. Fit pairwise transformations for a broad set of language pairs including unrelated ones (Romance languages + Mandarin, Swahili, Japanese, Hindi). Identify language-ID dimensions via PCA on all pairwise difference vectors. Run tree-recovery at every projection dimensionality from 0 to k. Report the full accuracy-vs-projection curve. This characterizes separability of the phylogenetic signal from language-ID without any arbitrary projection threshold.

Phase 2: Cross-family replication. Repeat the full pipeline on Germanic (English, German, Dutch, Swedish, Danish) and Dravidian (Tamil, Kannada, Telugu, Malayalam). For Dravidian, use the Dravidian Etymological Dictionary (DEDR, Burrow and Emeneau) as the cognate source, restricted to high-confidence entries. For each family, report tree-recovery accuracy, layer profile, and the projection curve. Run on at least three model architectures with different training compositions: a general multilingual model, an Indic-focused model (Sarvam or IndicLLaMA), and one additional architecture. Cross-model consistency is part of the evidence.

Phase 3: Areal-outgroup test. For each family, include a contact language and see where tree-fitting places it. Marathi for Dravidian, Maltese for Romance, Finnish for Germanic (Uralic, but with centuries of Scandinavian contact). Correct outgroup placement is the positive prediction. Separately, test typologically similar but genealogically unrelated pairs to confirm that composition error does not produce spurious phylogenetic grouping.

Phase 4: Documentation-confound control. Replicate the pipeline on language families with sparse digital reconstruction literature. If tree recovery succeeds for an under-documented Bantu or Austronesian subfamily at comparable accuracy to Romance, retrieval of comparative scholarship is probably not the mechanism. If accuracy drops sharply, the retrieval explanation gains support, and the strong claim is scaled back. This phase is reported honestly either way.

Phase 5: Signal decomposition. Decompose the phylogenetically informative transformation components (those that survive language-ID projection and contribute to correct tree recovery) into principal or sparse components. Ask whether the leading components correspond to known morphological paradigm differences, phonological correspondences, or syntactic parameter settings between the languages. This characterizes what the signal contains, contingent on Phases 1-4 establishing that it exists. Follow-up analysis, not the central test.

Phase 6 (optional): Mechanistic localization. With the structural result in hand, identify which layers and attention heads mediate the phylogenetically informative transformations. Sparse autoencoders come in here, if at all, because at this point we know what we are looking for.

Success criteria and falsification

The pilot has a binary outcome: correct Romance tree topology or not. If not, the project ends. Beyond the pilot, success is graded. Minimum publishable result: correct tree recovery for Romance and at least one additional family, across at least two model architectures, with the projection curve showing a separability peak. Strong result: correct recovery for all three families, correct outgroup placement for contact languages, and survival on a documentation-sparse family. Preregistered falsification threshold: if the correct topology is not in the top 3 out of 15 for any family on any model, that family is counted as a failure. If all families fail on a given model, that model lacks phylogenetic structure. All results, including negatives, are reported.

What Rigor Looks Like Here

Three model families minimum with documented differences in multilingual training composition. Strict disjoint train/test splits for all transformation fitting. Full projection curve reported, not a single projection choice. Preregistered before evaluation: the tree-recovery procedure, the falsification threshold, and the areal-outgroup predictions. Tokenization controls for script-distinct languages (Dravidian). Cognate sets restricted to high-confidence entries, with sensitivity analysis on broader inclusion reported alongside. Cross-family replication is not optional. Documentation-confound control is not optional. Negative results are headlines, not footnotes. Code and elicitation protocols open-sourced at submission.

SECTION 4: THE SO WHAT

Depth Level of Your Design

Level 0 — We propose a structural geometry exists. *Yes, but framed as measurement, not proposal.*

Level 1 — Representation geometry recovers correct phylogenetic tree topologies. *Core claim. Reached.*

Level 2 — The signal is separable from language-ID (projection curve), specific to genealogy not areal contact (outgroup test), and robust to retrieval confounds (documentation-sparse replication). *Reached.*

Level 3 — General principle: neural networks trained on related instances of a structured domain discover phylogenetic transformation structure from distributional data alone. Language families are the first clean test case. *Aim for this.*

Impact Type

Primarily methodological: composition-error-minimizing tree recovery is a new technique for extracting relational structure from learned representations, applicable outside linguistics. Conceptual: if positive, the finding changes what we think multilingual pretraining produces. Cross-lingual geometry is not just language identification plus typological overlay; it includes separable phylogenetic structure. Practical: for historical linguists working on low-resource families, a new data source for phylogenetic inference where traditional comparative data is sparse.

Broadest Truthful Audience

Computational phylogeneticists across domains (biology, linguistics, cultural evolution) who want new tree-inference methods. ML researchers interested in what relational structure shows up in learned representations without supervision over that structure. A journalist could write: researchers found that language models, trained only to predict the next word, have organized their internal representations into a family tree that matches what historical linguists spent a century reconstructing by hand.

The Science Version

Title: Phylogenetic Tree Recovery from Representation Geometry in Multilingual Language Models

Abstract lead: We show that the tree topology minimizing composition error of linear transformations between related-language representations in multilingual LLMs recovers established phylogenies for Romance, Germanic, and Dravidian. The phylogenetic signal is separable from language identification, specific to genealogy rather than areal contact, and consistent across model architectures. These results show that multilingual pretraining encodes phylogenetic branching structure as a measurable geometric property of the representation space.